

Data Mining

Classification: Alternative Techniques

Bayesian Classifiers

Classification

- Bayes Classification Methods 
- Model Evaluation and Selection
- Techniques to Improve Classification Accuracy:
Ensemble Methods
- Summary

Bayesian Classification: Why?

- A statistical classifier: performs *probabilistic prediction, i.e.*, predicts class membership probabilities

Bayesian Classification: Why?

- A statistical classifier: performs *probabilistic prediction, i.e.*, predicts class membership probabilities
- Foundation: Based on Bayes' Theorem.
- Performance: A **simple** Bayesian classifier, *naïve Bayesian classifier*, **has comparable performance** with decision tree and neural network classifiers

Bayes' Theorem: Basics

$$P(B) = \sum_{i=1}^M P(B|A_i)P(A_i)$$

$$P(H|\mathbf{X}) = \frac{P(\mathbf{X}|H)P(H)}{P(\mathbf{X})} = P(\mathbf{X}|H) \times P(H) / P(\mathbf{X})$$

Bayes' Theorem: Basics

- Total probability Theorem: $P(B) = \sum_{i=1}^M P(B|A_i)P(A_i)$
- Bayes' Theorem: $P(H | \mathbf{X}) = \frac{P(\mathbf{X} | H)P(H)}{P(\mathbf{X})} = P(\mathbf{X} | H) \times P(H) / P(\mathbf{X})$
 - Let $\mathbf{X}$ be a data sample ("evidence"): class label is unknown
 - Let H be a *hypothesis* that X belongs to class C

Bayes' Theorem: Basics

- Total probability Theorem: $P(B) = \sum_{i=1}^M P(B|A_i)P(A_i)$
- Bayes' Theorem:
$$P(H | \mathbf{X}) = \frac{P(\mathbf{X} | H)P(H)}{P(\mathbf{X})} = P(\mathbf{X} | H) \times P(H) / P(\mathbf{X})$$
 - Let $\mathbf{X}$ be a data sample ("evidence"): class label is unknown
 - Let H be a *hypothesis* that $\mathbf{X}$ belongs to class C
 - Classification is to determine $P(H|\mathbf{X})$, (i.e., *posteriori probability*): the probability that the hypothesis holds given the observed data sample $\mathbf{X}$

Bayes' Theorem: Basics

- Total probability Theorem: $P(B) = \sum_{i=1}^M P(B|A_i)P(A_i)$
- Bayes' Theorem: $P(H | \mathbf{X}) = \frac{P(\mathbf{X} | H)P(H)}{P(\mathbf{X})} = P(\mathbf{X} | H) \times P(H) / P(\mathbf{X})$
 - Let $\mathbf{X}$ be a data sample ("evidence"): class label is unknown
 - Let H be a *hypothesis* that $\mathbf{X}$ belongs to class C
 - Classification is to determine $P(H|\mathbf{X})$, (i.e., *posteriori probability*): the probability that the hypothesis holds given the observed data sample $\mathbf{X}$
 - $P(H)$ (*prior probability*): the initial probability
 - E.g., $\mathbf{X}$ will buy computer, regardless of age, income, ...
 - $P(\mathbf{X})$: probability that sample data is observed
 - $P(\mathbf{X}|H)$ (*likelihood*): the probability of observing the sample $\mathbf{X}$, given that the hypothesis holds

Bayes' Theorem: Basics

- Total probability Theorem: $P(B) = \sum_{i=1}^M P(B|A_i)P(A_i)$
- Bayes' Theorem: $P(H | \mathbf{X}) = \frac{P(\mathbf{X} | H)P(H)}{P(\mathbf{X})} = P(\mathbf{X} | H) \times P(H) / P(\mathbf{X})$
 - Let $\mathbf{X}$ be a data sample ("evidence"): class label is unknown
 - Let H be a *hypothesis* that $\mathbf{X}$ belongs to class C
 - Classification is to determine $P(H|\mathbf{X})$, (i.e., *posteriori probability*): the probability that the hypothesis holds given the observed data sample $\mathbf{X}$
 - $P(H)$ (*prior probability*): the initial probability
 - E.g., $\mathbf{X}$ will buy computer, regardless of age, income, ...
 - $P(\mathbf{X})$: probability that sample data is observed
 - $P(\mathbf{X}|H)$ (*likelihood*): the probability of observing the sample $\mathbf{X}$, given that the hypothesis holds
 - E.g., Given that $\mathbf{X}$ will buy computer, the prob. that $\mathbf{X}$ is 31..40, medium income

Classification Is to Derive the Maximum Posteriori

- Let D be a training set of tuples and their associated class labels, and each tuple is represented by an n -D attribute vector $\mathbf{X} = (x_1, x_2, \dots, x_n)$
- Suppose there are m classes $C_1, C_2, \dots, C_m$.
- Classification is to derive the maximum posteriori, i.e., the maximal $P(C_i|\mathbf{X})$
- This can be derived from Bayes' theorem

$$P(C_i|\mathbf{X}) = \frac{P(\mathbf{X}|C_i)P(C_i)}{P(\mathbf{X})}$$

- Since $P(\mathbf{X})$ is constant for all classes, only

$$P(C_i|\mathbf{X}) = P(\mathbf{X}|C_i)P(C_i)$$

needs to be maximized

Naïve Bayes Classifier: Training Dataset

Class:

C1:buys_computer = 'yes'

C2:buys_computer = 'no'

Data to be classified:

X = (age $\leq$ 30,

Income = medium,

Student = yes

Credit_rating = Fair)

age	income	student	credit_rating	com
$\leq$ 30	high	no	fair	no
$\leq$ 30	high	no	excellent	no
31...40	high	no	fair	yes
$>$ 40	medium	no	fair	yes
$>$ 40	low	yes	fair	yes
$>$ 40	low	yes	excellent	no
31...40	low	yes	excellent	yes
$\leq$ 30	medium	no	fair	no
$\leq$ 30	low	yes	fair	yes
$>$ 40	medium	yes	fair	yes
$\leq$ 30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
$>$ 40	medium	no	excellent	no

Naïve Bayes Classifier: An Example

- $P(C_i)$: $P(\text{buys_computer} = \text{"yes"}) = 9/14 = 0.643$
 $P(\text{buys_computer} = \text{"no"}) = 5/14 = 0.357$
- Compute $P(\text{X}|C_i)$ for each class

age	income	student	credit rating	com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31..40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31..40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31..40	medium	no	excellent	yes
31..40	high	yes	fair	yes
>40	medium	no	excellent	no

Naïve Bayes Classifier: An Example

$P(C_i)$: $P(\text{buys_computer} = \text{"yes"}) = 9/14 = 0.643$

$P(\text{buys_computer} = \text{"no"}) = 5/14 = 0.357$

Compute $P(X|C_i)$ for each class

$P(\text{age} = \text{"<=30"} \mid \text{buys_computer} = \text{"yes"}) = 2/9 = 0.222$

age	income	student	credit_rating	com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31..40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31..40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31..40	medium	no	excellent	yes
31..40	high	yes	fair	yes
>40	medium	no	excellent	no

Naïve Bayes Classifier: An Example

- $P(C_i)$: $P(\text{buys_computer} = \text{"yes"}) = 9/14 = 0.643$
 $P(\text{buys_computer} = \text{"no"}) = 5/14 = 0.357$

- Compute $P(X|C_i)$ for each class

$$P(\text{age} = \text{"<=30"} \mid \text{buys_computer} = \text{"yes"}) = 2/9 = 0.222$$

$$P(\text{age} = \text{"<= 30"} \mid \text{buys_computer} = \text{"no"}) = 3/5 = 0.6$$

age	income	student	credit rating	com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31..40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31..40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31..40	medium	no	excellent	yes
31..40	high	yes	fair	yes
>40	medium	no	excellent	no

Naïve Bayes Classifier: An Example

- $P(C_i)$: $P(\text{buys_computer} = \text{"yes"}) = 9/14 = 0.643$
 $P(\text{buys_computer} = \text{"no"}) = 5/14 = 0.357$

- Compute $P(X|C_i)$ for each class

$$P(\text{age} = \text{"<=30"} \mid \text{buys_computer} = \text{"yes"}) = 2/9 = 0.222$$

$$P(\text{age} = \text{"<= 30"} \mid \text{buys_computer} = \text{"no"}) = 3/5 = 0.6$$

$$P(\text{income} = \text{"medium"} \mid \text{buys_computer} = \text{"yes"}) = 4/9 = 0.444$$

$$P(\text{income} = \text{"medium"} \mid \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$$

$$P(\text{student} = \text{"yes"} \mid \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$$

$$P(\text{student} = \text{"yes"} \mid \text{buys_computer} = \text{"no"}) = 1/5 = 0.2$$

$$P(\text{credit_rating} = \text{"fair"} \mid \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$$

$$P(\text{credit_rating} = \text{"fair"} \mid \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$$

- $X = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$

age	income	student	credit_rating	com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31..40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31..40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31..40	medium	no	excellent	yes
31..40	high	yes	fair	yes
>40	medium	no	excellent	no

Naïve Bayes Classifier: An Example

- $P(C_i)$: $P(\text{buys_computer} = \text{"yes"}) = 9/14 = 0.643$
 $P(\text{buys_computer} = \text{"no"}) = 5/14 = 0.357$

- Compute $P(X|C_i)$ for each class

$$P(\text{age} = \text{"<=30"} \mid \text{buys_computer} = \text{"yes"}) = 2/9 = 0.222$$

$$P(\text{age} = \text{"<= 30"} \mid \text{buys_computer} = \text{"no"}) = 3/5 = 0.6$$

$$P(\text{income} = \text{"medium"} \mid \text{buys_computer} = \text{"yes"}) = 4/9 = 0.444$$

$$P(\text{income} = \text{"medium"} \mid \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$$

$$P(\text{student} = \text{"yes"} \mid \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$$

$$P(\text{student} = \text{"yes"} \mid \text{buys_computer} = \text{"no"}) = 1/5 = 0.2$$

$$P(\text{credit_rating} = \text{"fair"} \mid \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$$

$$P(\text{credit_rating} = \text{"fair"} \mid \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$$

- $X = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$

$$P(X|C_i) : P(X|\text{buys_computer} = \text{"yes"}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = 0.044$$

age	income	student	credit_rating	com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31-40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31-40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31-40	medium	no	excellent	yes
31-40	high	yes	fair	yes
>40	medium	no	excellent	no

Naïve Bayes Classifier: An Example

age	income	student	credit_rating	com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31-40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31-40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
<=30	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31-40	medium	no	excellent	yes
31-40	high	yes	fair	yes
>40	medium	no	excellent	no

- $P(C_i): P(\text{buys_computer} = \text{"yes"}) = 9/14 = 0.643$
 $P(\text{buys_computer} = \text{"no"}) = 5/14 = 0.357$
- Compute $P(X|C_i)$ for each class
 $P(\text{age} = \text{"<=30"} | \text{buys_computer} = \text{"yes"}) = 2/9 = 0.222$
 $P(\text{age} = \text{"<= 30"} | \text{buys_computer} = \text{"no"}) = 3/5 = 0.6$
 $P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"yes"}) = 4/9 = 0.444$
 $P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$
 $P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$
 $P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"no"}) = 1/5 = 0.2$
 $P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$
 $P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$
- $X = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$
 $P(X|C_i): P(X|\text{buys_computer} = \text{"yes"}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = 0.044$
 $P(X|\text{buys_computer} = \text{"no"}) = 0.6 \times 0.4 \times 0.2 \times 0.4 = 0.019$
 $P(X|C_i) \cdot P(C_i): P(X|\text{buys_computer} = \text{"yes"}) \cdot P(\text{buys_computer} = \text{"yes"}) = 0.028$

Naïve Bayes Classifier: An Example

age	income	student	credit_rating	com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31..40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31..40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31..40	medium	no	excellent	yes
31..40	high	yes	fair	yes
>40	medium	no	excellent	no

- $P(C_i): P(\text{buys_computer} = \text{"yes"}) = 9/14 = 0.643$

$$P(\text{buys_computer} = \text{"no"}) = 5/14 = 0.357$$

- Compute $P(X|C_i)$ for each class

$$P(\text{age} = \text{"<=30"} | \text{buys_computer} = \text{"yes"}) = 2/9 = 0.222$$

$$P(\text{age} = \text{"<= 30"} | \text{buys_computer} = \text{"no"}) = 3/5 = 0.6$$

$$P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"yes"}) = 4/9 = 0.444$$

$$P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$$

$$P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$$

$$P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"no"}) = 1/5 = 0.2$$

$$P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$$

$$P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$$

- $X = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$

$$P(X|C_i): P(X|\text{buys_computer} = \text{"yes"}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = 0.044$$

$$P(X|\text{buys_computer} = \text{"no"}) = 0.6 \times 0.4 \times 0.2 \times 0.4 = 0.019$$

$$P(X|C_i) \cdot P(C_i): P(X|\text{buys_computer} = \text{"yes"}) \cdot P(\text{buys_computer} = \text{"yes"}) = 0.028$$

$$P(X|\text{buys_computer} = \text{"no"}) \cdot P(\text{buys_computer} = \text{"no"}) = 0.007$$

Naïve Bayes Classifier: An Example

age	income	student	credit	rating	com
<=30	high	no	fair	no	no
<=30	high	no	excellent	no	no
31..40	high	no	fair	yes	yes
>40	medium	no	fair	yes	yes
>40	low	yes	fair	yes	yes
>40	low	yes	excellent	no	no
31..40	low	yes	excellent	yes	yes
<=30	medium	no	fair	no	no
<=30	low	yes	fair	yes	yes
<=30	medium	yes	fair	yes	yes
<=30	medium	yes	excellent	yes	yes
31..40	medium	no	excellent	yes	yes
31..40	high	yes	fair	yes	yes
>40	medium	no	excellent	no	no

- $P(C_i): P(\text{buys_computer} = \text{"yes"}) = 9/14 = 0.643$

$$P(\text{buys_computer} = \text{"no"}) = 5/14 = 0.357$$

- Compute $P(X|C_i)$ for each class

$$P(\text{age} = \text{"<=30"} | \text{buys_computer} = \text{"yes"}) = 2/9 = 0.222$$

$$P(\text{age} = \text{"<= 30"} | \text{buys_computer} = \text{"no"}) = 3/5 = 0.6$$

$$P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"yes"}) = 4/9 = 0.444$$

$$P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$$

$$P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$$

$$P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"no"}) = 1/5 = 0.2$$

$$P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$$

$$P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$$

- $X = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$

$$P(X|C_i): P(X|\text{buys_computer} = \text{"yes"}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = 0.044$$

$$P(X|\text{buys_computer} = \text{"no"}) = 0.6 \times 0.4 \times 0.2 \times 0.4 = 0.019$$

$$P(X|C_i) \cdot P(C_i): P(X|\text{buys_computer} = \text{"yes"}) \cdot P(\text{buys_computer} = \text{"yes"}) = 0.028$$

$$P(X|\text{buys_computer} = \text{"no"}) \cdot P(\text{buys_computer} = \text{"no"}) = 0.007$$

Therefore, X belongs to class ("buys_computer = yes")

Classification Is to Derive the Maximum Posteriori

- Let D be a training set of tuples and their associated class labels, and each tuple is represented by an n -D attribute vector $\mathbf{X} = (x_1, x_2, \dots, x_n)$
- Suppose there are m classes $C_1, C_2, \dots, C_m$.
- Classification is to derive the maximum posteriori, i.e., the maximal $P(C_i|\mathbf{X})$
- This can be derived from Bayes' theorem

$$P(C_i|\mathbf{X}) = \frac{P(\mathbf{X}|C_i)P(C_i)}{P(\mathbf{X})}$$

- Since $P(\mathbf{X})$ is constant for all classes, only

$$P(C_i|\mathbf{X}) = P(\mathbf{X}|C_i)P(C_i)$$

needs to be maximized

Naïve Bayes Classifier: Comments

- Advantages
 - Easy to implement
 - Good results obtained in most of the cases
- Disadvantages
 - Assumption: class conditional independence, therefore loss of accuracy
 - Practically, dependencies exist among variables
 - E.g., hospitals: patients: Profile: age, family history, etc.
Symptoms: fever, cough etc., Disease: lung cancer, diabetes, etc.
 - Dependencies among these cannot be modeled by Naïve Bayes Classifier
- How to deal with these dependencies? Bayesian Belief Networks

Chapter 8. Classification: Basic Concepts

- Bayes Classification Methods
- Model Evaluation and Selection 
- Techniques to Improve Classification Accuracy:
Ensemble Methods
- Summary

Assessment Methods for Classification

Accuracy

classifier accuracy: predicting class label

Speed

time to construct the model (training time)

time to use the model (classification/prediction time)

Robustness: handling noise and missing values

Scalability: efficiency in disk-resident databases

Interpretability

understanding and insight provided by the model

Accuracy of Classification Models

■ Confusion Matrix

	PREDICTED CLASS		
		Class=Yes	Class=No
ACTUAL CLASS	Class=Yes	a (TP)	b (FN)
	Class=No	c (FP)	d (TN)

Metrics for Performance Evaluation...

		PREDICTED CLASS	
		Class=Yes	Class=No
ACTUAL CLASS	Class=Yes	a (TP)	b (FN)
	Class=No	c (FP)	d (TN)

- Most widely-used metric:

$$\text{Accuracy} = \frac{a + d}{a + b + c + d} = \frac{TP + TN}{TP + TN + FP + FN}$$

-
- **Precision quantifies** the number of positive class predictions that actually belong to the positive class.
 - **Recall quantifies** the number of positive class predictions made out of all positive examples in the dataset.
 - **F-Measure** provides a single score that balances both the concerns of precision and recall in one number.

Classifier Evaluation Metrics: Precision and Recall, and F-measures

- **Precision:** exactness – what % of tuples that the classifier labeled as positive are actually positive

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\text{recall} = \frac{TP}{TP + FN}$$

$$F = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

Classifier Evaluation Metrics: Precision and Recall, and F-measures

- **Precision:** exactness – what % of tuples that the classifier labeled as positive are actually positive

$$\textit{precision} = \frac{TP}{TP + FP}$$

- **Recall:** completeness – what % of positive tuples did the classifier label as positive?

$$\textit{recall} = \frac{TP}{TP + FN}$$

Classifier Evaluation Metrics:

Precision and Recall, and F-measures

- **Precision:** exactness – what % of tuples that the classifier labeled as positive are actually positive

$$\text{precision} = \frac{TP}{TP + FP}$$

- **Recall:** completeness – what % of positive tuples did the classifier label as positive?

$$\text{recall} = \frac{TP}{TP + FN}$$

$$F = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

Classifier Evaluation Metrics: Precision and Recall, and F-measures

- **Precision**: exactness – what % of tuples that the classifier labeled as positive are actually positive

$$\text{precision} = \frac{TP}{TP + FP}$$

- **Recall**: completeness – what % of positive tuples did the classifier label as positive?

$$\text{recall} = \frac{TP}{TP + FN}$$

- Perfect score is 1.0
- Inverse relationship between precision & recall
- **F measure (F_1 or F-score)**: harmonic mean of precision and recall,

$$F = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

Classifier Evaluation Metrics: Example

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\text{recall} = \frac{TP}{TP + FN}$$

Actual Class \ Predicted class	cancer = yes	cancer = no	Total	Recognition(%)
cancer = yes	90	210	300	30.00 (sensitivity)
cancer = no	140	9560	9700	98.56 (specificity)
Total	230	9770	10000	

Classifier Evaluation Metrics: Example

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\text{recall} = \frac{TP}{TP + FN}$$

Actual Class \ Predicted class	cancer = yes	cancer = no	Total	Recognition(%)
cancer = yes	90	210	300	30.00 (sensitivity)
cancer = no	140	9560	9700	98.56 (specificity)
Total	230	9770	10000	96.40 (accuracy)

- **Precision** = $90/230 = 39.13\%$
- **Recall** = $90/300 = 30.00\%$
- **F-Score** = $2 * P * R / (P + R) = 2 * 39.13 * 30 / (39.13 + 30) = 33.96$
- **Accuracy** = $(90 + 9560) / 10000$

Classifier Evaluation Metrics: Confusion Matrix

Confusion Matrix:

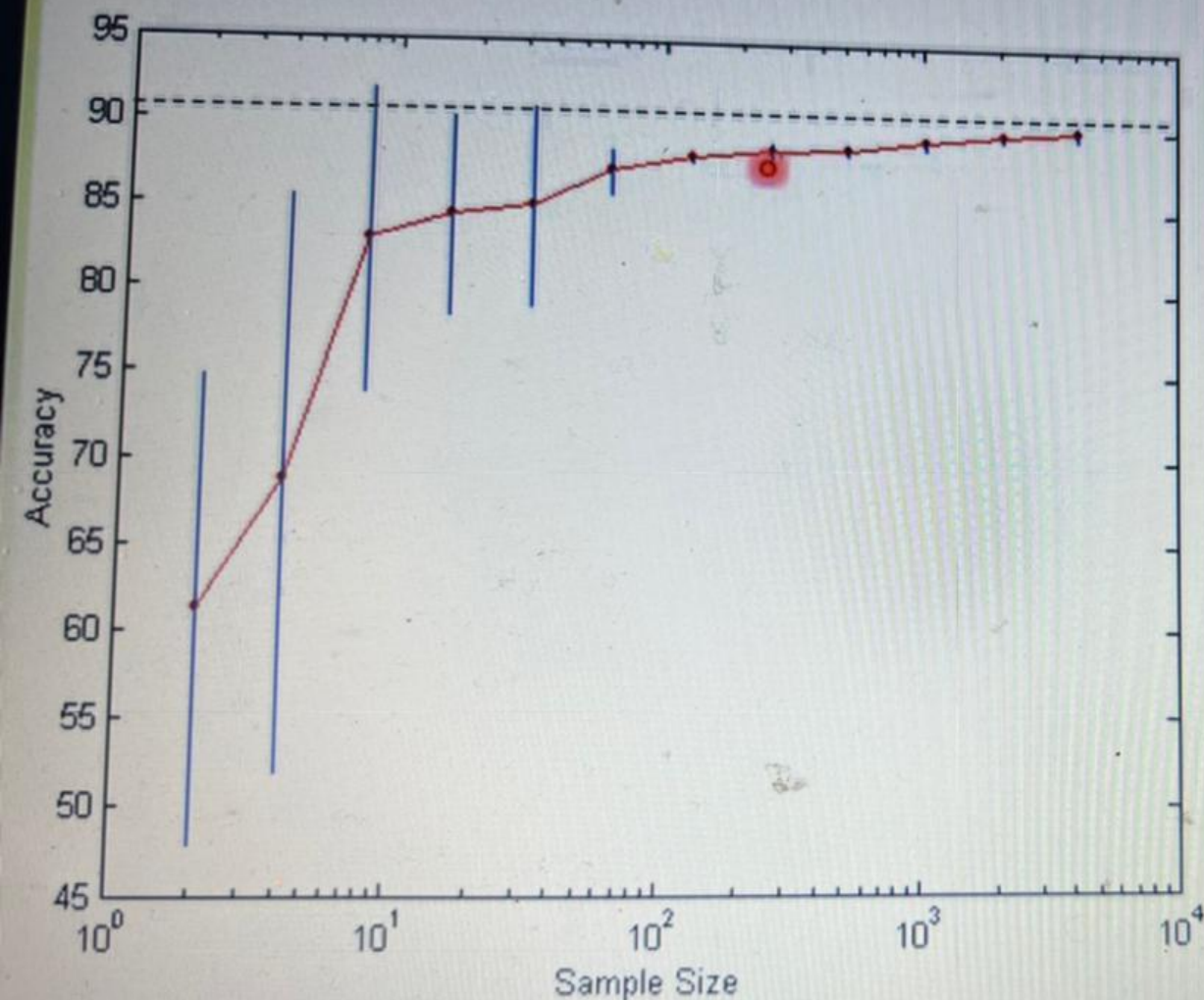
Actual class \ Predicted class	C_1	$\neg C_1$
C_1	True Positives (TP)	False Negatives (FN)
$\neg C_1$	False Positives (FP)	True Negatives (TN)

Example of Confusion Matrix:

Actual class \ Predicted class	buy_computer = yes	buy_computer = no	Total
buy_computer = yes	6954	46	7000
buy_computer = no	412	2588	3000
Total	7366	2634	10000

- **Accuracy:** $\text{Accuracy} = (TP + TN) / \text{All} = (6954 + 2588) / 10000 =$
- **Precision** $= (TP) / (TP + FP) = 6954 / (7366) =$
- **Recall** $= TP / (TP + FN) = 6954 / 7000 =$

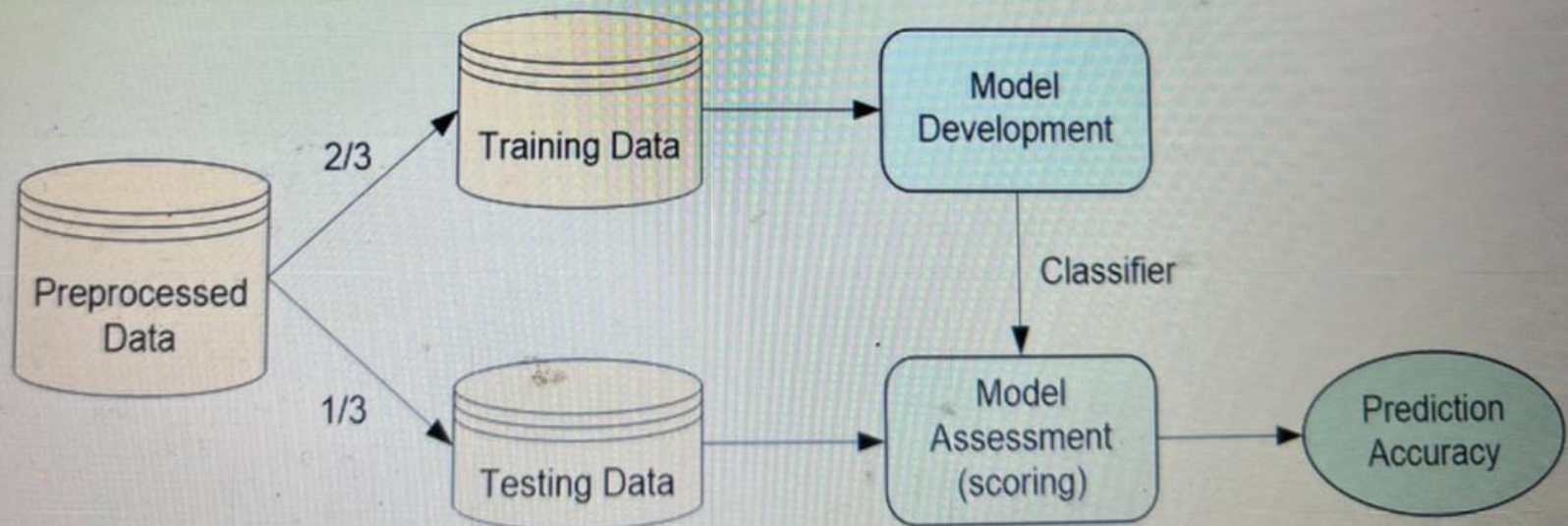
Learning Curve



- Learning curve shows how accuracy changes with varying sample size
- Requires a sampling schedule for creating learning curve

Estimation Methodologies for Classification

- **Simple split** (or holdout or test sample estimation)
 - Split the data into 2 mutually exclusive sets training (~70%) and testing (30%)

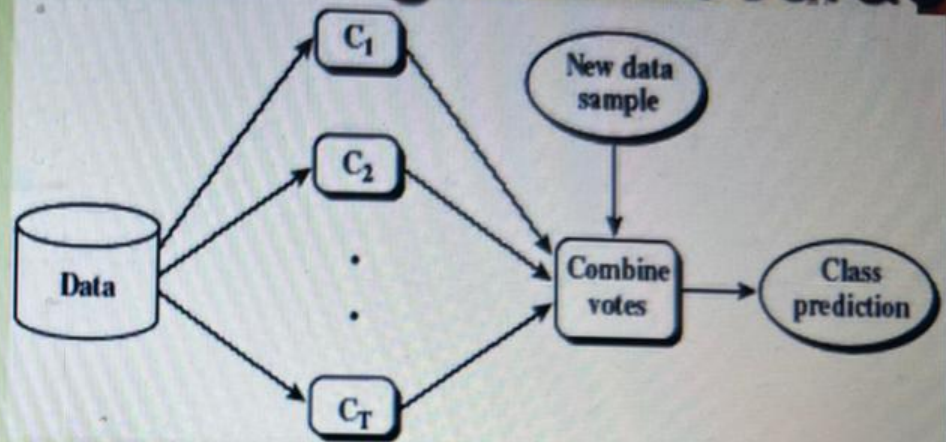


Chapter 8. Classification: Basic Concepts

- Bayes Classification Methods
- Model Evaluation and Selection
- Techniques to Improve Classification Accuracy:
Ensemble Methods
- Summary



Ensemble Methods: Increasing the Accuracy



■ Ensemble methods

- Use a combination of models to increase accuracy
- Combine a series of k learned models, $M_1, M_2, \dots, M_k$, with the aim of creating an improved model M^*

■ Popular ensemble methods

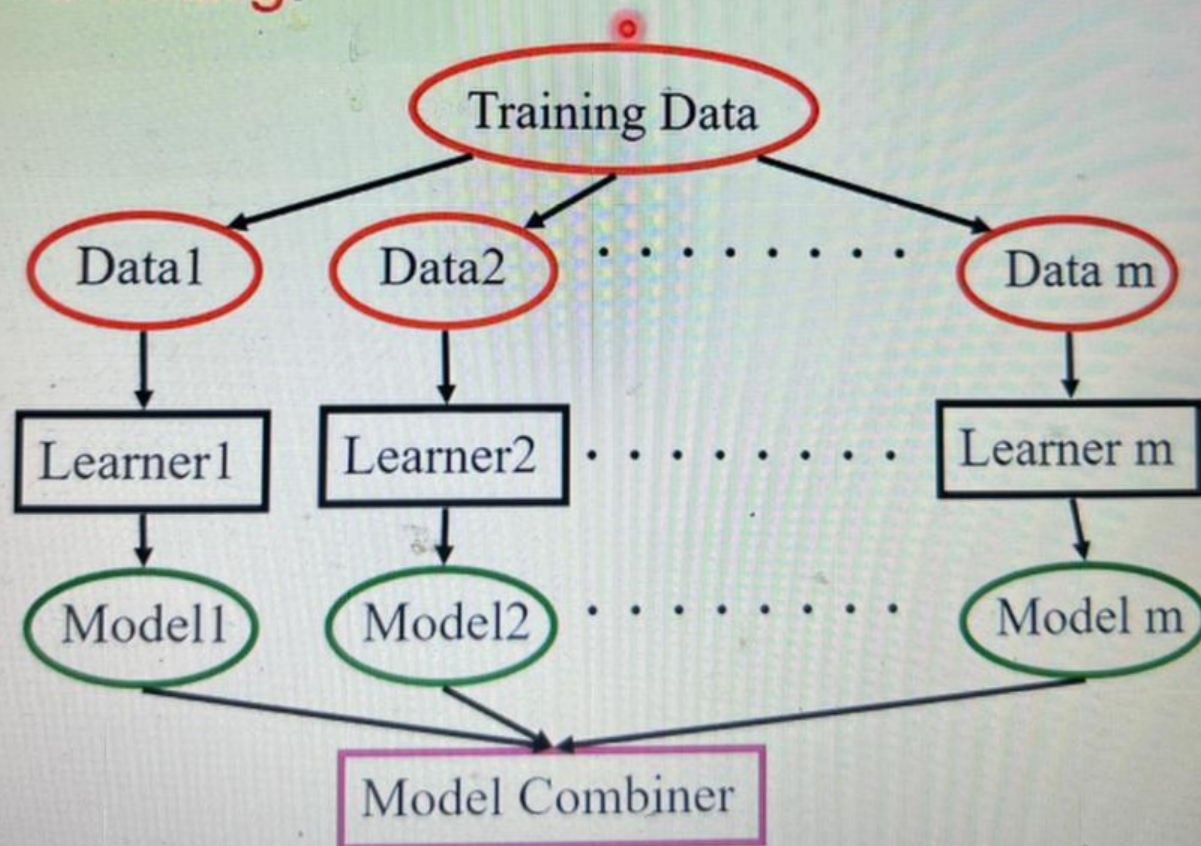
- **Bagging:** averaging the prediction over a collection of classifiers
- **Boosting:** weighted vote with a collection of classifiers

Increasing Classifier accuracy

- Based on using the outputs from several classifiers:
- **Bagging** : Taking the majority vote. (**Analogy**: Diagnosis based on multiple doctors' majority vote given test tuple)
- **Accuracy**
 - Often significantly better than a single classifier
 - For noise data: not considerably worse, more robust
 - Proved improved accuracy in prediction
- **Boosting** : Combining all the outcomes based on the weights of each classifier. Weights are assigned according to the classifier accuracy.

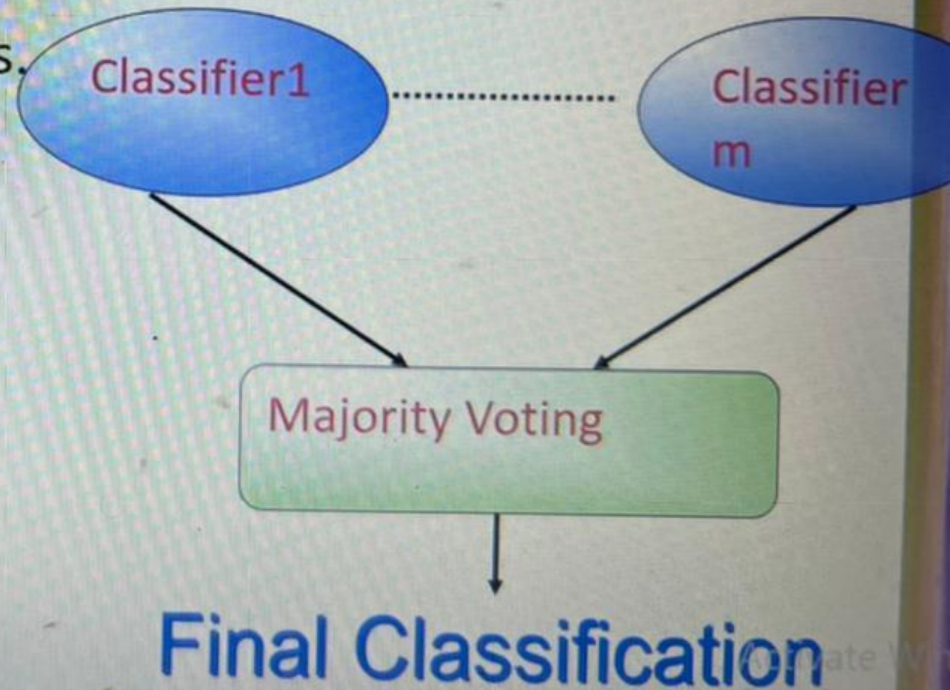
Learning Ensembles

- Learn multiple alternative definitions of a concept using **different training data** or **different learning algorithms**.
- **Combine decisions** of multiple definitions, e.g. using **weighted voting**.



Voting

- ❑ Voting denotes the simplest method of combining predictions from multiple classifiers.
- ❑ In its simplest form, called majority voting.
- ❑ Each Classifier contributes a single vote (its own classification).
- ❑ The collective prediction is decided by majority of the votes.



Example: Weather Forecast

Reality							
1							
2							
3							
4							
5							
Combine							

Example: Weather Forecast

Reality							
1							
2							
3							
4							
5							
Combine							

Classification of Class-Imbalanced Data Sets

- Class-imbalance problem: Rare positive example but numerous negative ones, e.g., medical diagnosis, fraud, oil-spill, fault, etc.
- Traditional methods assume a balanced distribution of classes and equal error costs: not suitable for class-imbalanced data

Classification of Class-Imbalanced Data Sets

- Class-imbalance problem: Rare positive example but numerous negative ones, e.g., medical diagnosis, fraud, oil-spill, fault, etc.
- Traditional methods assume a balanced distribution of classes and equal error costs: not suitable for class-imbalanced data
- **Typical methods for imbalance data in 2-class classification:**
 - **Oversampling:** re-sampling of data from positive class
 - **Under-sampling:** randomly eliminate tuples from negative class

Summary

- Classification is a form of data analysis that extracts models describing important data classes.
- Effective and scalable methods have been developed for decision tree induction, Naive Bayesian classification, and many other classification methods.
- Evaluation metrics include: accuracy, sensitivity, specificity, precision, recall, F measure, and F_β measure.
- Stratified k-fold cross-validation is recommended for accuracy estimation.
- Bagging and boosting can be used to increase overall accuracy by learning and combining a series of individual models.
- No single method has been found to be superior over all others for all data sets

Building a Training Dataset

Example : Predicting Good or Bad credit

Predict the credit behavior of a credit card applicant from applicant's attributes:

- personal status
- job type
- housing type
- savings account

These are all categorical variables; better suited to Naïve Bayesian classifier than to logistic regression.

personal_status	job	housing	savings_status	credit_class
male single	skilled	own	no known savings	good
female div/dep/mar	skilled	own	<100	bad
male single	unskilled resident	own	<100	good
male single	skilled	for free	<100	good
male single	skilled	for free	<100	bad
male single	unskilled resident	for free	no known savings	good
male single	skilled	own	500<=X<1000	good
male single	high qualif/self emp/mgm	rent	<100	good
male div/sep	unskilled resident	own	>=1000	good
male mar/wid	high qualif/self emp/mgm	own	<100	bad
female div/dep/mar	skilled	rent	<100	bad
female div/dep/mar	skilled	rent	<100	bad
female div/dep/mar	skilled	own	<100	good
male single	unskilled resident	own	<100	bad
female div/dep/mar	skilled	rent	<100	good
female div/dep/mar	unskilled resident	own	100<=X<500	bad
male single	skilled	own	no known savings	good
male single	skilled	own	no known savings	good
female div/dep/mar	high qualif/self emp/mgm	for free	<100	bad
male single	skilled	own	500<=X<1000	good
male single	skilled	own	<100	good
male single	skilled	rent	500<=X<1000	good
male single	unskilled resident	rent	<100	good
male single	skilled	own	100<=X<500	good
male mar/wid	skilled	own	no known savings	good
male single	unskilled resident	own	<100	good
male mar/wid	unskilled resident	own	<100	good

Technical Description - Bayes' Law

$$P(B|A) = \frac{P(A \wedge B)}{P(A)} = \frac{P(A|B)P(B)}{P(A)}$$

- B is the class label:
 - $B \in \{b_1, b_2, \dots, b_n\}$
- A is the specific assignment of input v
 - $A = (a_1, a_2, \dots, a_m)$



Reverend Thomas Bayes

Building a Naïve Bayesian Classifier

- To build a Naïve Bayesian classifier, collect the following statistics from the training data:
 - $P(b_j)$ for all the class labels.
 - $P(a_i | b_j)$ for all possible assignments of the input variables and class labels.

Credit example:

- class labels: {good, bad}
 - $P(\text{good}) = 0.7$
 - $P(\text{bad}) = 0.3$
- aggregates for housing
 - $P(\text{own} | \text{bad}) = 0.62$
 - $P(\text{own} | \text{good}) = 0.75$
 - $P(\text{rent} | \text{bad}) = 0.23$
 - $P(\text{rent} | \text{good}) = 0.14$
 - ... and so on

Back to Credit Example

Credit Example: X

- female
- owns home
- Self-employed
- savings > \$1000

$P(\text{good}|X) > P(\text{bad}|X)$:
Assign X the label
"good"

$$P(\text{good}|X) \sim (0.28 \cdot 0.75 \cdot 0.14 \cdot 0.06) \cdot 0.7 = 0.0012$$

$$P(\text{bad}|X) \sim (0.36 \cdot 0.62 \cdot 0.17 \cdot 0.02) \cdot 0.3 = 0.0002$$

a_i	b_j	$P(a_i b_j)$
female	good	0.28
female	bad	0.36
own	good	0.75
own	bad	0.62
self emp	good	0.14
self emp	bad	0.17
savings>1K	good	0.06
savings>1K	bad	0.02